SolCacao Artisanal Chocolate Platform

Database Schema Specification — MongoDB

Engine: MongoDB 7+ | ODM: Mongoose | Modeling: embedded subdocuments + ObjectId references

users

Retail and B2B account records. B2B accounts carry an embedded corporateProfile subdocument.

Field	Type	Notes
_id	ObjectId	Primary key
firstName	String	required
lastName	String	required
email	String	required, unique, indexed
passwordHash	String	required
accountType	String	enum: 'retail' | 'b2b', default 'retail'
phone	String	optional
preferredLocale	String	default 'en'
addresses	[Address]	embedded array (see Address subdocument)
corporateProfile	CorporateProfile	embedded, present only if accountType = 'b2b'
createdAt / updatedAt	Date	timestamps

categories

Bilingual product categories; self-referencing for sub-category hierarchies.

Field	Type	Notes
_id	ObjectId	Primary key
nameEn / nameEs	String	required
slug	String	required, unique, indexed
parentCategoryId	ObjectId	ref -> categories, nullable

products

Catalog items: single-origin bars, truffle collections, and seasonal chocolates.

Field	Type	Notes
_id	ObjectId	Primary key
nameEn / nameEs	String	required
descriptionEn / descriptionEs	String	optional

Field	Type	Notes
sku	String	required, unique, indexed
cacaoPercentage	Number	required
origin	String	required
productType	String	enum: 'bar' | 'truffle' | 'seasonal'
price	Number (Decimal128)	required
imageUrl	String	optional
categoryId	ObjectId	ref -> categories, nullable
isActive	Boolean	default true
inventory	Inventory	embedded (see Inventory subdocument)
createdAt	Date	timestamp

carts

Active shopping carts for both guests (session id) and logged-in users; merged on login.

Field	Type	Notes
_id	ObjectId	Primary key
userId	ObjectId	ref -> users, nullable (guest cart)
guestSessionId	String	nullable
status	String	enum: 'active' | 'converted' | 'abandoned'
items	[CartItem]	embedded array (see CartItem subdocument)
createdAt / updatedAt	Date	timestamps

orders

Completed D2C retail orders. Shipping address and line items are embedded as a point-in-time snapshot.

Field	Type	Notes
_id	ObjectId	Primary key
userId	ObjectId	ref -> users, required
orderNumber	String	required, unique, indexed
totalAmount	Number (Decimal128)	required
status	String	default 'pending'
shippingAddress	Address	embedded snapshot (see Address subdocument)
items	[OrderItem]	embedded array (see OrderItem subdocument)
placedAt	Date	timestamp

b2bQuoteRequests

Submissions from the 4-step B2B customization wizard, including uploaded brand assets.

Field	Type	Notes
_id	ObjectId	Primary key
corporatePartnerUserId	ObjectId	ref -> users, required
cacaoProfile	String	required
packagingPreference	String	optional
estimatedQuantity	Number	optional
status	String	enum: 'submitted' | 'in_review' | 'quoted' | 'completed'
notes	String	optional
files	[QuoteFile]	embedded array (see QuoteFile subdocument)
createdAt	Date	timestamp

contactInquiries

General contact / support / wholesale inquiry submissions routed via email.

Field	Type	Notes
_id	ObjectId	Primary key
name	String	required
email	String	required
category	String	enum: 'general' | 'wholesale' | 'support'
message	String	required
status	String	default 'new'
createdAt	Date	timestamp

customSamplers

User-built custom chocolate sampler configurations.

Field	Type	Notes
_id	ObjectId	Primary key
userId	ObjectId	ref -> users, nullable
guestSessionId	String	nullable
samplerName	String	optional
items	[SamplerItem]	embedded array (see SamplerItem subdocument)
createdAt	Date	timestamp

Embedded Subdocument Shapes

These are not standalone collections — they exist only nested inside a parent document (no independent _id required, though Mongoose adds one by default for array subdocuments).

Address

Field	Type	Notes
label	String	e.g. 'Home', 'Office'
addressLine1	String	required
addressLine2	String	optional
city	String	required
state	String	optional
zipCode	String	required
country	String	required
isDefault	Boolean	default false

CorporateProfile

Field	Type	Notes
companyName	String	required
taxId	String	optional
industry	String	optional
contactPerson	String	optional
verificationStatus	String	enum: 'pending' | 'verified' | 'rejected'

Inventory

Field	Type	Notes
stockQuantity	Number	default 0
reorderThreshold	Number	default 10
updatedAt	Date	timestamp

CartItem

Field	Type	Notes
productId	ObjectId	ref -> products
quantity	Number	default 1
addedAt	Date	timestamp

OrderItem

Field	Type	Notes
productId	ObjectId	ref -> products
quantity	Number	required
unitPrice	Number (Decimal128)	price captured at time of purchase

QuoteFile

Field	Type	Notes
fileUrl	String	required
fileType	String	optional
originalFilename	String	optional
uploadedAt	Date	timestamp

SamplerItem

Field	Type	Notes
productId	ObjectId	ref -> products
quantity	Number	default 1

References (Cross-Collection ObjectId Links)

From Collection	References	Field & Notes
carts	users	userId → optional (guest carts have none)
orders	users	userId → required
b2bQuoteRequests	users	corporatePartnerUserId → required
customSamplers	users	userId → optional (guest samplers have none)
products	categories	categoryId → optional
categories	categories	parentCategoryId → optional (self-reference)

Embedded Relationships

Parent Collection	Field	Subdocument	Notes
users	addresses	[Address]	One user embeds multiple saved addresses.
users	corporateProfile	CorporateProfile	One B2B user embeds exactly one corporate profile.
products	inventory	Inventory	One product embeds exactly one inventory record.
carts	items	[CartItem]	One cart embeds multiple line items.
orders	shippingAddress	Address	One order embeds one address snapshot.
orders	items	[OrderItem]	One order embeds multiple line items with captured price.
b2bQuoteRequests	files	[QuoteFile]	One quote request embeds multiple uploaded files.
customSamplers	items	[SamplerItem]	One sampler embeds multiple product selections.

Note: ES/EN localization of static UI strings (navigation, labels, form text) is served via static JSON dictionaries, not a database collection.